1. Problem Statement

Many users create weak and predictable passwords, which makes their accounts vulnerable to hacking and unauthorized access. Weak passwords can be easily guessed or cracked by attackers.

2. Objective

The main objective of this project is to develop a system that helps users create strong and secure passwords. It also checks the strength of passwords and spreads awareness about cybersecurity.

3. Proposed Solution

This project provides a simple system where users can enter their password to check its strength. The system analyzes the password based on different security rules such as length, use of uppercase letters, numbers, and special characters. It also generates strong passwords for users.

4. Features

- Password strength checker
- Random password generator
- User-friendly interface
- Security awareness

5. Working of Project

In this system, the user enters a password. The system checks the password strength based on various parameters like:

- Length of password
- Use of uppercase letters
- Use of numbers
- Use of special characters

Based on these factors, the system classifies the password as Weak, Medium, or Strong.

It also generates a secure password suggestion for better safety.

6. Technologies Used

Python Programming Language

VS Code

Basic libraries (random, string)

7. Applications

Used in login systems

Helps in securing user accounts

Useful in cybersecurity awareness

8. Future Scope

AI-based password suggestions

Integration with websites and apps

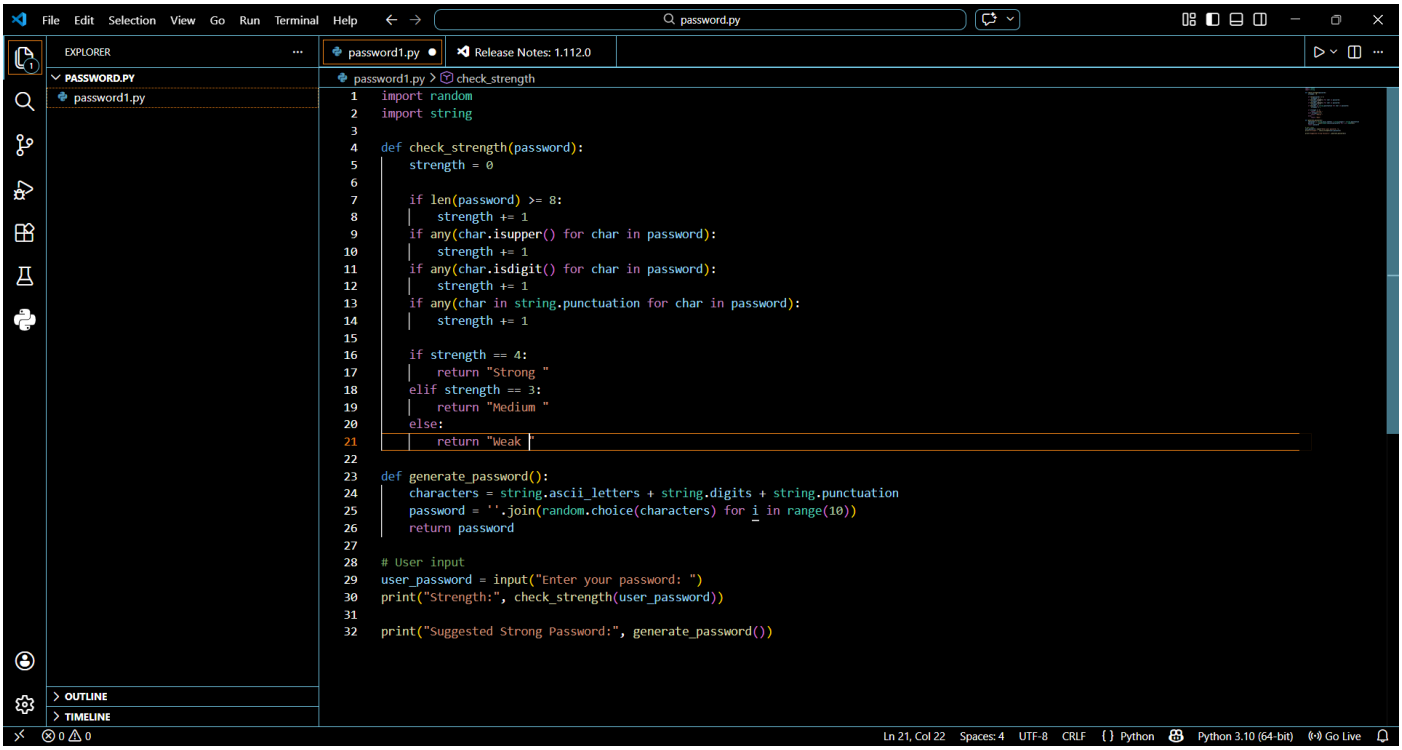
Multi-language support

9. Conclusion

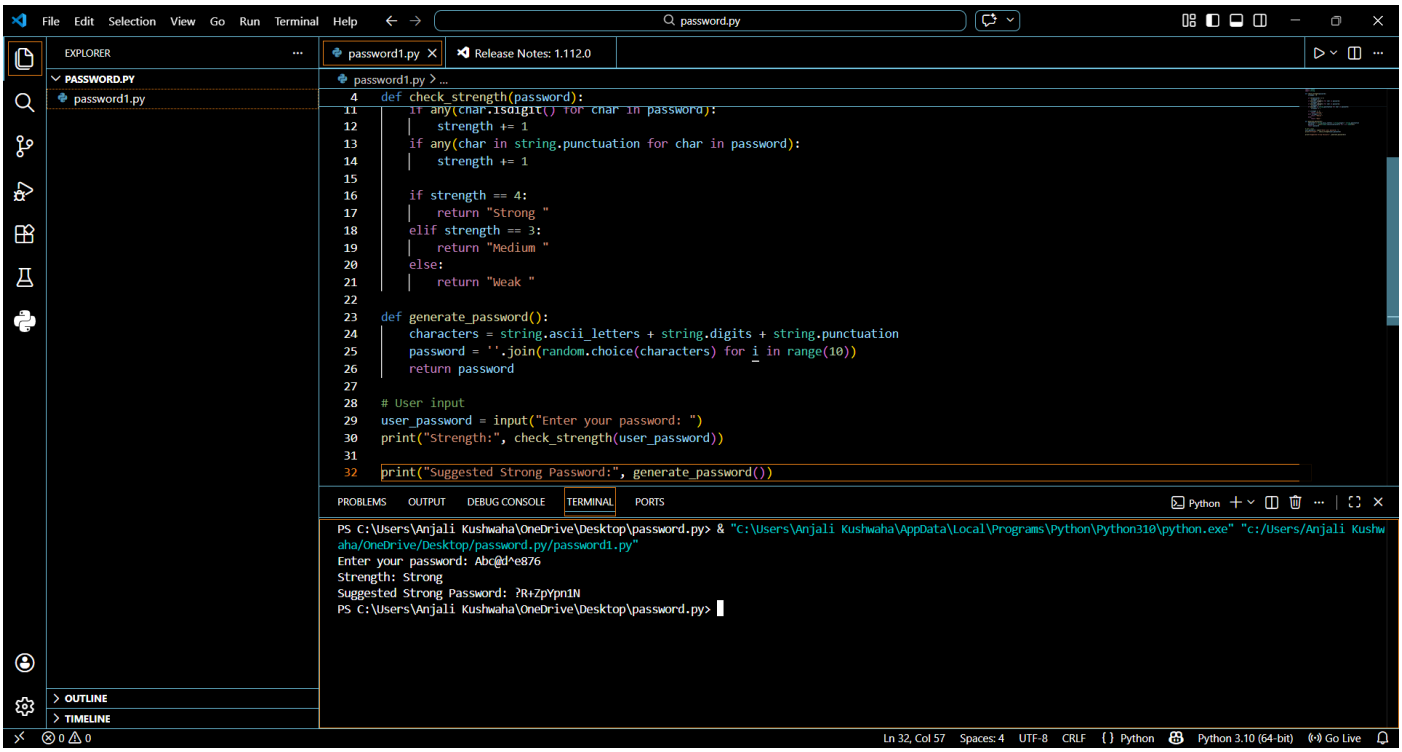
This project helps users understand the importance of strong passwords and improves their online security. It provides a simple and effective way to generate and check secure passwords.

10. Screenshot

Code:



Output:


password1.py
def check_strength(password):
if any(char.isdigit() for char in password):
strength += 1
if any(char in string.punctuation for char in password):
strength += 1
if strength == 4:
return "Strong "
elif strength == 3:
return "Medium "
else:
return "Weak "
def generate_password():
characters = string.ascii_letters + string.digits + string.punctuation
password = ''.join(random.choice(characters) for i in range(10))
return password
# User input
user_password = input("Enter your password: ")
print("Strength:", check_strength(user_password))
print("Suggested Strong Password:", generate_password())
PS C:\Users\Anjali Kushwaha\OneDrive\Desktop\password.py> & "C:\Users\Anjali Kushwaha\AppData\Local\Programs\Python\Python310\python.exe" "c:/Users/Anjali Kushwaha/OneDrive/Desktop/password.py/password1.py"
Enter your password: Abc@d^e876
Strength: Strong
Suggested Strong Password: ?R+ZpYpn1N
PS C:\Users\Anjali Kushwaha\OneDrive\Desktop\password.py>